

Business Process Continuous Improvement System Based on Workflow Mining Technology

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Abstract

As an important part of business process management, continuous improvement is a crucial factor determining the enterprises' capability of keeping competition in rapidly changing marketplace. In this paper, we integrate the data mining technology into the workflow management system to find out the tacit business process knowledge, and put forward the concept of Workflow Mining. Based on this foundation, business process continuous improvement framework on workflow historical process information mining is proposed. It includes four levels: historical information acquisition, performance evaluation, structural defects identification and generation of improved model. Key techniques of each layer are discussed, and prototype system is developed to support the implementation of this method. Enterprise application shows that this system has strong flexibility and extensibility, and its structure is reasonable.

1. Introduction

Process improvement is a critical phase of business process management. It aims to design new business process to ensure the enterprises keeping competition capacity in rapidly changing marketplace^[1]. From the point view of influence scope and duration of this action, process improvement can be divided into two categories, which are Business Process Reengineering (BPR) and Continuous Process Improvement (CPI). CPI is often understood as *the planned, organized and systematic process of ongoing, incremental and company-wide change of existing practices aimed at improving company performance*^[2]. It now becomes the hotspot of this research area because of its low risks, easiness to conduct and continuity of its effects.

As a key technique for business process management, workflow management system integrates the activities' execution with the information transmission and resource allocation strategies, and

now, business performance evaluation and improvement based on workflow technique attracts more attentions not only in theories, but also in practices. Much of the initial literature takes the theories of Petri-net, stochastic, and/or queuing to the workflow models, and gets dynamic performance indicator values by simulation of the process^[3, 4]. Although these methods can get the result quickly by low cost, it is only the stimulating execution of the process, which can not reflect the real situation of the business. In [5], Koksai P. pointed out that process instance information logged in history repository provides a solid foundation for business process monitoring, analysis and reengineering, and intelligent analysis of the workflow audit data will add flexibility and adaptability to the workflow management systems.

This paper contributes to integrating data mining technique into the workflow management system, and proposed the general framework of continuous process improvement based on workflow mining technology. On this basis, related key techniques are discussed.

2. Framework of continuous improvement

Workflow process model is the regular expression of business process – describing the business process activities and their executing orders. In the traditional approach, business process is mapped into the initial workflow model which is suitable for the marketplace, and automated by workflow engines. But with the changes of surrounding environment, activity participants or process monitor may adjust the execution of process instances by supervisory API, e.g. add a new activity or skip the execution of an activity, to improve the quality or efficiency of the process. It should not be seen as “exception” when occurs frequently because it may reflect the initiative of employees to adjust their work to the surrounding environment. So, intelligent analysis on the workflow history repository can determine the effectiveness and efficiency of business processes, and point out the improvement directions of the processes^[6, 7].

In the past, the goal of workflow mining is to collect data at runtime to support workflow design and reverse the process. But in this research, we refer the workflow mining technology as the combination of data mining and workflow management system from all aspects. It takes the workflow audit repository as data source, and aims at discovering the tacit knowledge not only for process model design, but also for process evaluation and improvement.

Here, we propose the general framework of continuous process improvement based on workflow mining technology, and carry out our research work on this framework.

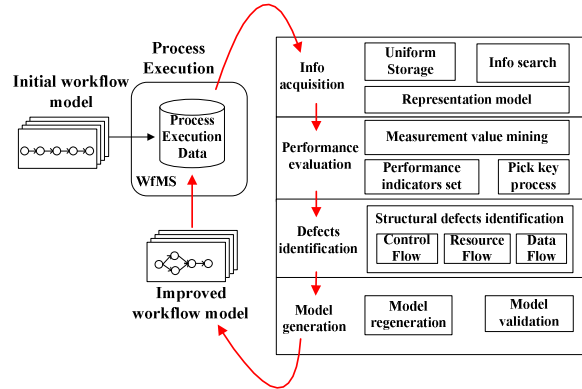


Fig.1 Continuous improvement framework

This framework includes four layers: information acquisition layer, performance evaluation layer, structural defects identification layer and improved model generation layer. Historical process information from heterogeneous workflow management systems is organized into a uniformed XML file in the first layer. It provides the data foundation for CPI. In the performance evaluation layer, business process execution performance is measured, and improvement activity is launched when necessary. Structural defects identification layer aims to find out the direction of process improvement. It provides process managers with toolkits which can discover process structural defects from activity composition aspect, i.e., control flow, data flow and resource flow. Model generation layer is to reverse the process model from the audit repository and verify its correctness. Output of this layer is the improved process model, and replaces the original model to be parsed and executed in the engine. This framework reflects the idea of Deming loop in process management, and lays the architecture basis for continuous process improvement.

3. Key techniques

3.1 Information acquisition technique

Information acquisition layer is to collect historical process instance information from heterogeneous workflow audit repositories. Reasonable representation method, rapid collection mechanism, and uniformed storage format of process instance information provides data foundation for continuous improvement.

From the point view of workflow, process measurement toolkit may offer some evaluation functions according to four dimensions, i.e., process definition, process instance, activity instance and activity resource. These four measurement objects are organized in hierarchy format in concept domain, and we can model this relationship in tree structure. Fig.2 depicts the business process model of an aircraft plant from the workflow aspect.

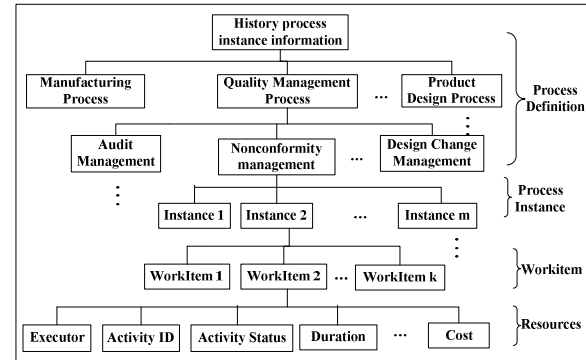


Fig. 2 Model case of process instance information

Extensible Markup Language (XML) is a general-purpose specification for creating custom mark-up languages. It allows users to define the mark-up elements with tree structure. With the support of the process information model, execution logs from heterogeneous workflow management systems can be standardized into an XML file to provide uniform data source for performance analysis.

3.2 Performance evaluation technique

Performance measurement system is to determine the set of metrics used to quantify both the efficiency and effectiveness of actions, and detect bottleneck, deadlock, and other incorrect facts in processes. Bad performance measurement system cannot find out process problems in time, and cannot meet with the “in-time and active” requirement of CPI.

Time to market (T), quality (Q), cost (C), service (S), and environment (E) are five fundamental attributes to businesses’ survival and development.

And among them, TQC are three most critical ones because they relate closely with the business logistics, information flow and capital flows. In [8], Hui-ping L discussed the detailed indicators from these three aspects, such as process turn-around time, activity waiting time, activity executing cost, and et al. Introducing these indicators into workflow systems, Xin-yu L proposed a workflow performance analysis net (WPAN) model and designed a mining algorithm to evaluate the process execution performance [9]. And here, we develop our toolkit based on these researches.

3.3 Structural defects identification technique

In previous methodologies and tools, statistic results of final output are used as the data source to analyze process conformance. But in workflow systems, “activity” is the atomic object which integrates data and resource with the process. A process is composed of a group of interact-each-other activities, and efficiency/quality of one activity may impact the execution of another one. So process structural defects identification from activity composition aspect is critical for process improvement.

As method above, process instance log can be divided into two subsets based on the average criteria values. Mining and comparing the frequent sequential patterns contained in the two subsets can find out the structure which determines the process inferior from control flow aspect. Aiming at treating with the loop structure in process, we put forward an improved graph pattern mining approach based on Apriori algorithm in [10], where a frequent pattern of size k , denoted as G_i , can joined with another frequent pattern of size k , denoted as G_j , to form a candidate frequent pattern of size $(k+1)$, if there exists a source vertex s in G_i and a sink vertex e in G_j such that the graph derived from G_i by subtracting node s equals with the graph derived from G_j by subtracting node e . This coalition operator only considers the source vertexes and sink vertexes, which reduces the amount of redundant frequent candidates generated in traditional Apriori iterations.

In [7], Van der Aalst put forwards an approach, defines metrics, and presents a tool to mine social networks from workflow audit repository. This method can help administrators to uncover the interaction pattern of activity participant in the process, and here we take this method to detect the interaction structural defects from participant resource.

3.4 Process model regeneration technique

The ultimate goal of CPI based on workflow system is to generate the workflow model which meets with

the market circumstances. In traditional approaches, process administrators may adjust the initial process model occasionally or frequently by hand to meet the circumstance changing. This work costs highly and is error-prone because the business processes and the workflow modeling tools are both complex to grasp. So how to design an effective algorithm to reconstruct a process model compliant with the well-executed workflow instance is so important.

In paper [11], we put forward a heuristic algorithm to mine the control flow of workflow model, where activity execution is modeled as an interval, and algorithm logic is designed based on the relationship between two activity execution intervals, such as *disjoint*, *contained*, and *overlapped*. This method reduces the time complexity compared with Agrawal algorithm, and can regenerate the model with complex loop structures effectively.

4. Case study

Here, we take the quality management processes of an aircraft manufacture plants as an example to develop our prototype system.

Now, there are a plenty of workflow management systems applied in practice. In order to improve the scalability of our system, we adopt the plug-in technique to develop our information acquisition toolkit. Fig. 3 depicts the sketch of this toolkit.

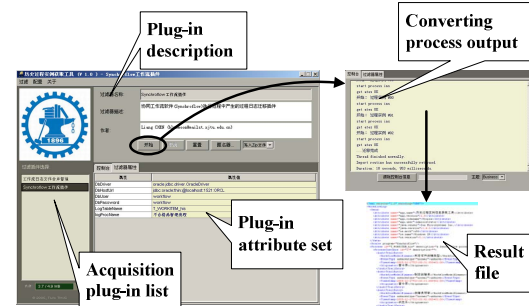


Fig.3 Historical process instance acquisition tool

Information acquisition toolkit collects the process instance audit data from various workflow management systems, and converts them into a uniform XML file. Using this file as the data source, process performance evaluation tool is developed based on C/S architecture. Fig. 4 shows the main structure of this tool.

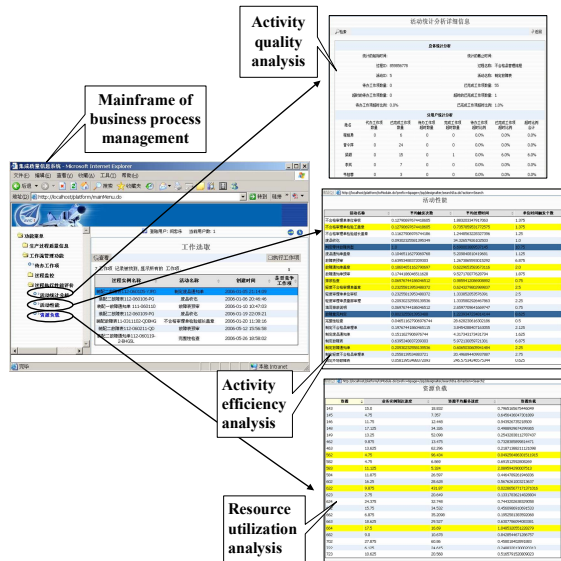


Fig. 4 Process performance evaluation tool

When the critical indicator values can not satisfy the requirement of process execution, we can use the structural defects identification tools to find out improvement directions. Here, frequent activity sequential pattern mining and participant social networks identification tools are developed as Fig. 5.

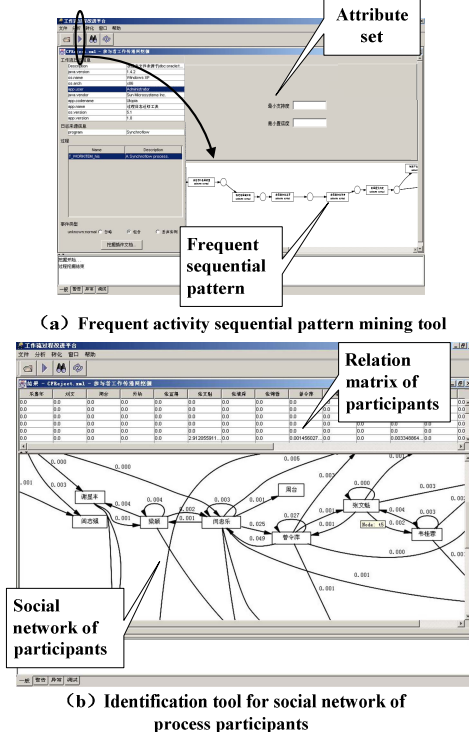


Fig. 5 Structural defects identification tool

Fig. 6 shows the operation process of process regeneration tool. In this tool, we take the algorithm mentioned in section 3.4 to reverse control flow of process model (described as XML file). By graphical

modeling tool, we can modify the reversed control flow and add needed resources to the activities.

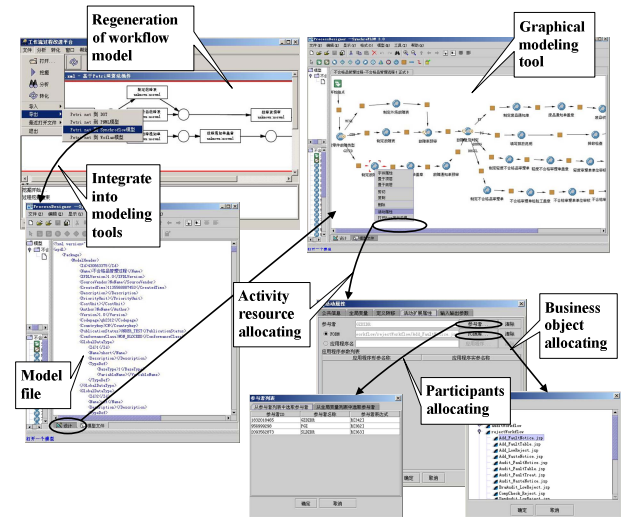


Fig. 6 Process model regeneration and integration tool

The new model is generated from the filtered process instance information which reflects the real market circumstances, and can be regarded as improved process model. This model then replaces the initial model to be parsed and executed in the workflow engine, and launches the new round of improvement.

5. Conclusion

In this paper, we integrate the data mining technique into workflow management systems, and present the research framework of business process continuous improvement based on workflow mining technology. Key techniques from the four aspects are digested, such as process information acquisition, process execution performance evaluation, process structural defects identification and process model regeneration. Implementation of this method meets with the Deming loops of process management, and can help business to achieve the goal of continuous process improvement. Based on this framework, prototype system is developed to support the implementation of this methodology.

6. Acknowledgment

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